3.1 Self-Principled Critique Pairwise GenRM for Creative Writing

The Self-Principled Critique Tuning (SPCT) framework, proposed by Liu et al. (2025), offers a method for unified scoring of single or multiple responses via textual critique. Building upon this, we introduce our **Pairwise Writing GenRM**, which adapts and refines the SPCT methodology specifically for the nuances of creative writing. Our approach incorporates two key modifications to enhance its suitability and effectiveness for non-verifiable writing tasks:

- **Focused Application on Non-verifiable Writing:** Unlike the broader application of SPCT, our Pairwise Writing GenRM is exclusively applied to non-verifiable writing tasks. This focus stems from the observation that verifiable tasks, which possess definitive answers, can be more directly and reliably assessed using rule-based verifiable rewards.
- **Dedicated Pairwise Comparison:** We streamline the input format to solely consist of pairwise responses. This contrasts with SPCT's capability to handle single or multiple responses. We find that a dedicated pairwise setup simplifies the training process and yields more accurate comparative assessments by eliminating potential distractions from additional responses.

The development and training of our Pairwise Writing GenRM, based on the `Qwen3-32B-Base` model, follows a structured four-step pipeline:

1. **Data Filtering:** Selection of higher-quality data from raw preference datasets.
2. **SPCT Prompt Engineering:** Design of tailored SPCT prompts specifically for writing tasks.
3. **Cold-Start Fine-tuning:** Collection of a small set of cold-start data using a rejection sampling strategy to initially fine-tune the `Qwen3-32B-Base` model.
4. **RL-based Refinement:** Further enhancement of the GenRM's performance using rule-based reinforcement learning.

3.1.1 Data Filtering

Our GenRM training data originates from an in-house dataset of approximately 200K pairwise preferences, including 30K writing-related pairs. These writing pairs are re-scored using a scalar RM trained on the full preference set. We then filter for higher-quality pairs, characterized by a higher chosen response reward and a larger reward gap, resulting in approximately 10K original writing pairs. For GenRM evaluation, we use an in-house test set of 1K writing preferences. The final preference dataset can be described as:

$$\mathcal{D} = \{(x^{(i)}, y_c^{(i)}, y_r^{(i)})\}_{i=1}^{N} \tag{5}$$

where x is the query, y_c and y_r are the chosen and rejected responses.

3.1.2 Writing Principles and Critique

Our GenRM template, similar to SPCT, utilizes pre-defined general writing principles and model-generated specific principles (based on the input query and response pair) to form a detailed critique (Liu et al., 2025). The critique concludes with `\boxed{score_1, score_2}` to denote the comparative quality of the two responses.

The GenRM follows a sequential process: first generating specific principles, then the critique, and finally extracting the paired scores:

$$\{p_i\}_{i=1}^{m} \sim p_\theta\left(x, y_c, y_r\right), \quad \mathcal{C} \sim r_\theta\left(x, y_c, y_r, \{p_i\}_{i=1}^{m}\right), \quad \{S_c, S_r\} = (S_1, S_2) = f_{\text{extract}}(\mathcal{C}), \tag{6}$$

where p_θ denotes the principle generation function (sharing parameters θ with the reward generation function r_θ), which outputs specific principles $\{p_i\}$. $\mathcal{C}$ represents the detailed critique generated based on these principles. S_c and S_r are the individual quality scores for the chosen (y_c) and rejected (y_r) responses, respectively, corresponding to (S_1, S_2) based on their presentation order in the template. f_{extract} is the function to extract these paired scores from the critique $\mathcal{C}$. Notably, unlike SPCT, our GenRM outputs S_c and S_r as float numbers within the [0, 10] range, rather than integers.

3.1.3 Cold Start with RFT

To mitigate instability in the early stages of RL training and to enhance the granularity of specific principles, we perform a cold-start fine-tuning (RFT) of the initial RL actor. This involves curating a small, high-quality dataset as follows: First, we sample 1k writing preference pairs from the 10k higher-quality pairs in 3.1.1 and swap the positions of chosen and rejected responses to construct 2k SPCT prompts, with 50% (y_c, y_r) and 50% (y_r, y_c). Then we sample initial reasoning traces (specific principles and

critiques) for these SPCT prompts using `Claude-3.5-Sonnet`. Note that Claude's predictions exhibit a disproportional tendency (around 60%) to assign higher scores to the former response, suggesting a potential position bias problem.

Next, we employ a rejection process to refine the dataset. Specifically, for each SPCT prompt, we examine Claude's predicted $\{S_c, S_r\}$, the pair scores for the chosen (y_c) and rejected (y_r) responses respectively. If this prediction does not align with the known ground-truth preference for that pair (i.e., $S_c < S_r$), the prompt is discarded. We then proceed to only keep the original queries for which *both* of their corresponding SPCT prompt versions (i.e., the (y_c, y_r) version and the swapped (y_r, y_c) version) are both consistent with the ground-truth preference. As a result, we maintain 500 original writing preference pairs, doubling this set to 1,000 pairs by swapping the chosen and rejected responses. We fine-tune `Qwen3-32B-Base` with these 1k cold start data for one epoch.

3.1.4 Rule-based RL for Pairwise GenRM

The writing GenRM is further fine-tuned by GRPO (Shao et al., 2024), with rule-based outcome rewards. The predicted pair float scores $\{S_c, S_r\}$ are extracted and composed into two kinds of rewards: an accuracy reward and a format reward.

Accuracy Reward Formally, for the i-th output o_i in the group of G individual responses $\{o_i\}_{i=1}^{G}$, the accuracy reward is:

$$R_{\text{acc}} = \begin{cases} 1, & \text{if } S_c > S_r, \\ -1, & \text{if } S_c < S_r, \\ 0, & \text{otherwise,} \end{cases} \tag{7}$$

the rewards are computed for each response o_i, we omit the index i for notational simplicity.

Format Reward The format reward is also important, for both parsing process and numerical boundary:

$$R_{\text{format}} = \begin{cases} 1, & \text{if } 0 \leq S_c \leq 10 \text{ and } 0 \leq S_r \leq 10, \\ -1, & \text{otherwise.} \end{cases} \tag{8}$$

Score Margin As it is difficult for the writing GenRM to distinguish pair responses with fine-grained textual or semantic divergences, we introduce a heuristic weighting term to enhance GenRM's sensitivity to these differences:

$$R_{\text{acc}} = \begin{cases} R_{\text{acc}} \times \dfrac{(S_c - S_r)}{\tau_{\text{margin}}}, & \text{if } 0 < S_c - S_r < \tau_{\text{margin}}, \\ R_{\text{acc}}, & \text{otherwise,} \end{cases} \tag{9}$$

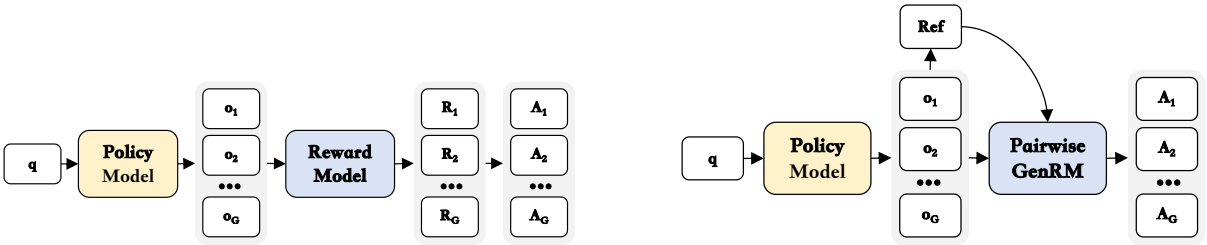
where the margin threshold is $\tau_{\text{margin}} = 2$ in our implementation.

Position Bias Weight Let P_{former} denote the probability that the GenRM prefers the first-presented (former) response in a pair. Ideally, we expect P_{former} to be 0.5 because the training data is deliberately balanced with respect to presentation order, as it includes swapped chosen-rejected response pairs. Furthermore, its variance (estimated from batch statistics) should approach 0 on this balanced training data. The reinforcement learning process helps converge the empirical estimate of P_{former} (i.e., the proportion of times the former response is preferred in a batch) to approximately 0.5, yet reducing the variance of these batch-wise proportions remains a challenge. To mitigate the variance of these batch-wise preference proportions, we apply a weighting to the advantage $\hat{A}_{i,t}$ that is dynamically calculated based on the preference distribution within the current batch:

$$\hat{A}_{i,t} = \begin{cases} \hat{A}_{i,t} \times \dfrac{\text{\#Latter Preference in Batch}}{\text{Batch Size}/2}, & \text{if } S_c > S_r \text{ and } y_c \text{ was presented first,} \\ \hat{A}_{i,t} \times \dfrac{\text{\#Former Preference in Batch}}{\text{Batch Size}/2}, & \text{if } S_c > S_r \text{ and } y_c \text{ was presented second.} \end{cases} \tag{10}$$

Dynamic Sampling We employed dynamic sampling following Yu et al. (2025). This technique involves over-sampling and then filtering out prompts for which the GenRM's predictive accuracy is 0 or 1.

3.2 RL with Bootstrapped Relative Policy Optimization



(a) Architecture of GRPO

(b) Architecture of BRPO

Figure 2: Demonstration of GRPO and our BRPO. BRPO implements bootstrap by randomly selecting reference response from the current group of policy responses, and achieves zero expectation for the final advantages directly.

In this stage, we introduce Bootstrapped Relative Policy Optimization (BRPO), an algorithm adapted from GRPO (Shao et al., 2024). BRPO is designed to utilize preference rewards obtained from the pairwise GenRM, facilitating robust learning for non-verifiable tasks. With BRPO, `Qwen3-32B-Base` is trained as a writing model directly, without supervised fine-tuning (SFT) as a preliminary step, termed `Writing Zero`. Furthermore, we apply BRPO to an in-house SFT model to create `Writing-R1`.

3.2.1 Bootstrapped Relative Policy Optimization

While Group Relative Policy Optimization (GRPO) (Shao et al., 2024) utilizes group-relative normalization of scalar rewards, defining such rewards reliably for non-verifiable tasks like creative writing is challenging due to their inherent subjectivity. Pairwise comparative assessment, however, offers a more robust and scalable evaluation method, aligning better with human judgments in these subjective domains. Bootstrapped Relative Policy Optimization (BRPO) is designed to leverage this by enabling dynamic, reference-free pairwise comparison. It achieves this by utilizing a bootstrapped response—randomly selected from the current group of policy rollouts—as a temporary reference for advantage estimation during RL training, as shown in Figure 2b. This core mechanism eliminates the need for fixed external references, allowing the policy model to continuously improve by learning from its own increasingly sophisticated outputs. BRPO consists of two main components: a novel *preference advantage* calculation and a *dynamic sampling* strategy.

For a given query, BRPO dynamically establishes a temporary reference by randomly selecting one response, o_{ref}, from the current group of G policy-generated responses $\{o_i\}_{i=1}^{G}$. This o_{ref} is then used for pairwise comparison against all responses in the group to obtain preference scores. In other words, BRPO employs a bootstrap approach to iteratively enhance its model capabilities by conducting multiple rounds of voting and calculating preference among its self-generated outputs. Consequently, our BRPO achieves zero expectation for advantage and requires no additional normalization.

Preference Advantage At the core of BRPO's advantage estimation is the direct use of binary pairwise preference signals, replacing the pointwise scalar normalization found in methods like GRPO (as seen in Equation 2). Specifically, for each output o_i within the group of G policy-generated responses $\{o_i\}_{i=1}^{G}$, its preference relative to the dynamically selected reference o_{ref} (as described previously) is determined. This results in a binary preference score R_i, obtained by comparing o_i and o_{ref} using the pairwise GenRM. The corresponding scores S_i (for o_i) and S_{ref} (for o_{ref}) are extracted from the GenRM's critique. The advantage $\hat{A}_{i,t}$ for the i-th response at token t is then directly set to this binary preference score:

$$R_i = \begin{cases} 1, & \text{if } S_i > S_{\text{ref}} \text{ in all voting conditions} \\ -1, & \text{otherwise} \end{cases}, \quad \text{and thus, } \hat{A}_{i,t} = R_i. \tag{11}$$

Dynamic Sampling BRPO also incorporates a dynamic sampling mechanism. As the policy model π_θ evolves during training, the quality of its generated responses will progress. A potential issue arises if a chosen o_{ref} for a particular query happens to be an outlier—either exceptionally strong or weak, or containing superficial patterns that the GenRM is overly sensitive to. In such cases, o_{ref} might be consistently judged as superior (or inferior) to nearly all other responses in the group $\{o_i\}_{i=1}^{G}$. This can lead to a batch of preference scores R_i that are predominantly all +1 or all -1, offering little discriminative signal for policy updates and potentially encouraging the policy to overfit to superficial patterns in